Akash Rai

+91 8097253859 | akashtooop@gmail.com | linkedin.com/in/akash-rai1701 | github.com/akashrai2003

Professional Summary

AI Engineer with 2+ years of hands-on experience building secure, focusing on both enterprise and local-first GenAI systems, optimizing LLM inference on NVIDIA and AMD hardware, and deploying production document-intelligence, RAG, agentic, and knowledge-graph platforms. Experienced in vLLM-based inference, Blackwell GPU optimization, backend and microservice architecture, air-gapped deployments, and end-to-end client delivery.

Technical Skills

Languages: Python, C++, TypeScript
LLM / ML: PyTorch, Hugging Face, LangChain, LangGraph, vLLM, PEFT/LoRA, sentence-transformers, CuPy
Backend / Data: FastAPI, Flask, gRPC, RabbitMQ, MongoDB, MySQL, Neo4j, SQLite, sqlite-vec, MinIO
Infrastructure: Docker, Docker Compose, Kubernetes, Azure, AWS, GitHub Actions, MLflow, Linux, CUDA

Experience

AI Engineer Feb 2025 – Present
Inferigence Quotient *Bengaluru, India*

- Architected and deployed an end-to-end **document intelligence platform for the Indian Navy**, evolving RAG-based report generation into a secure, on-premise local-LLM platform for document analysis, generation, operational suggestions, and decision support.
- Designed and deployed an **air-gapped microservices platform** with 5+ Dockerized services, RabbitMQ, MongoDB, MinIO, OCR pipelines, and vLLM inference; led multiple **multi-day on-site client deployments in Mumbai**, collaborating with stakeholders on requirements, production feedback, upgrades, and rollout in secure environments.
- Independently developed an **agentic geospatial assistant** supporting MBTiles-based map visualization, temporal reasoning, playback, and Vega-Lite-powered conversational analytics for interactive operational decision support.
- Built a **long-term memory knowledge base** that continuously ingests scanned documents through OCR, extracts timelines and operational entities, preserves supporting evidence, and revises previously stored facts when newer documents supersede earlier information.
- Optimized **Ministral-3-8B FP8** on NVIDIA RTX 5060 Ti (Blackwell) using FP8 KV cache and prompt-aware batching, extending context length from **22K to 40K** while serving batches of 10–20 requests at **∼750 output tokens/s** aggregate throughput using approximately 15 GB VRAM.
- Enabled **Gemma-4-12B NVFP4** inference on SM120 Blackwell hardware by resolving CUDA 12.9, FlashInfer, CUTLASS, JIT-cache, and kernel-build compatibility issues; containerized Qwen3-9B, Ministral-3-8B, and Gemma deployments using vLLM and Docker Compose.
- Designed an independent **GPU-accelerated gRPC inference service** by refactoring statistical and Random Forest classifiers into a CuPy-powered microservice, accelerating large-scale batch predictions while avoiding PyTorch/CUDA dependency bloat across downstream services.
- Conducted R&D on adapting local SLMs through **continued pre-training (DAPT) and PEFT/LoRA**; curated military-domain corpora from books and operational documents, implemented deduplication and preprocessing pipelines, trained models on **4× NVIDIA L40S GPUs** using AWS multi-GPU infrastructure, and benchmarked domain-specific performance against base-model baselines to evaluate deployment readiness.

Independent AI Engineer – Consulting Oct 2025 – May 2026
LeanAgile Nautics / Upwork *Remote*

- Owned the end-to-end architecture and development of an enterprise analytics assistant, replacing an inaccurate RAG-first design with **LLM-generated SAQL** over Salesforce Einstein Analytics to answer metric-heavy and numerical business questions correctly.
- Built routing, query generation, validation, execution, result formatting, fallback logic, document ingestion, and FastAPI endpoints for production use across enterprise logistics workflows.
- Delivered a second grounded knowledge system combining **Neo4j graph retrieval, vector embeddings, and semantic search** to answer questions across long finance, product, and marketing documents with evidence-backed context.

AI Developer Intern Sept 2024 – Feb 2025
AI Planet *Remote*

- Built a multi-agent assessment generator using **LangGraph state machines**, orchestrating LLM agents through a 7-stage workflow for job-aligned question generation and adaptive candidate evaluation.
- Developed modular components for **GenAI Stack**, a drag-and-drop LLM pipeline builder, while experimenting with agent architectures and domain-specific workflow design.

AI Intern Mar 2024 – Aug 2024
DeepLogicAI *Remote*

- Developed a **production-grade medical RAG pipeline** with FastAPI, replacing Vectara-based RAGaaS to reduce operational costs by **10x** while improving control over retrieval and deployment.
- Engineered a cascading retrieval pipeline combining dense retrieval, filtering, reranking, and context selection to balance recall, precision, and computational cost.
- Migrated large-scale document metadata from Pinecone to MySQL and designed an optimized indexing schema for faster filtering and improved query workflows.

Open Source Contributions

kepler.gl Contributor | Merged PR #3262 Dec 2025
- Fixed persistent LLM conversation state in the AI Assistant's restart flow in the **12K-star kepler.gl repository**, clearing internal library and Redux state to prevent stale OpenAI/Gemini history from being resent and wasting tokens.

ML-CaPsule – GSSoC'24 Contributor | Merged PR #750 Jun 2024
- Implemented Bayesian hyperparameter optimization using Gaussian Processes, TPE, and SMAC across MIT-BIH ECG and MNIST experiments, resolving issue #662 and reducing training time by **40%** through efficient hyperparameter sampling.

Selected Projects

Inkwell – Evidence-Backed AI Learning IDE | *React, TypeScript, Rust, Tauri, SQLite, vLLM, MCP* Jun 2026 – Present
- Built a local-first desktop application for technical PDFs using React 19, TypeScript, Rust, and Tauri 2, with PDFium/PDF.js rendering, background extraction, indexing, annotations, and durable SQLite/WAL storage.
- Designed a privacy-aware model-routing layer that defaults sensitive workflows to local OpenAI-compatible vLLM/Qwen models and gates optional cloud routing through explicit data-scope permissions, cancellation, timeouts, and concurrency controls.
- Implemented source-grounded AI actions using page text spans, exact PDF-space rectangles, and citation jumps; AI annotations are previewed as change sets and committed through atomic, auditable, reversible SQLite transactions with conflict protection and idempotent retries.
- Built personalized missions, Socratic assessments, review cards, evidence-linked learner memory, and a read-only MCP server exposing scoped document search and page-context tools without leaking local paths.

SocialGraph – LLM-Powered Knowledge Graph Pipeline | *FastAPI, Playwright, vLLM, NetworkX, MCP* 2026
- Architected a 5-stage asynchronous ETL pipeline that converts **1,500+ saved LinkedIn posts** into a navigable Obsidian knowledge graph, with idempotent checkpoints supporting crash-safe incremental reruns.
- Designed multi-provider LLM routing across self-hosted Qwen 9B FP8 on vLLM and Groq-hosted Llama 3.3 70B, with fallback chains for high-throughput classification, summarization, taxonomy synthesis, and semantic edge inference.
- Implemented Playwright/Voyager ingestion, URL enrichment, local embeddings, cosine-similarity edges, Leiden community detection, and NetworkX analytics; shipped a FastAPI visualization dashboard, **8-tool MCP server**, 25+ CLI commands, and 74+ tests with CI.

AMD AI Premier League – SFT Reasoning Agents | *Qwen3, PyTorch, torchtune, vLLM, MI300X* Jul 2025
- Shortlisted for AMD's AI hackathon and fine-tuned two Qwen3-4B SLMs for logic, mathematics, and puzzle-solving tasks, with team models evaluated in head-to-head benchmark rounds.
- Created task-focused reasoning data and SFT pipelines on AMD MI300X accelerators, then served the resulting models through vLLM for high-throughput evaluation.

Distributed LLM Fine-Tuning | *Horovod, Ray Tune, Azure, PEFT* May 2025
- Implemented distributed fine-tuning experiments on Azure CPU clusters using Horovod and Ray Tune, evaluating scalable training and hyperparameter-search workflows with PEFT.
- Benchmarked GGUF, AWQ, and GPTQ quantization approaches to compare model size, latency, and deployment trade-offs.

Selected Achievements

- Won **1st place at IEEE CAS REVA** (INR 30,000) and finished runner-up at an IEEE BMSCE technical competition.
- Placed in the **Top 10 of Intel Unnati** at the IIIT Hyderabad Mobility Summit and was selected for Amazon ML Summer School 2024.

Education

NMIT Bengaluru India
Bachelor of Engineering in Electronics and Communication; ***CGPA: 8.34*** *Dec 2021 – Jul 2025*